

JavaScript DOM Assignment: Selection and CRUD Operations

Objective

Practice the DOM concepts covered in class:

- Understanding the DOM
- Selecting elements using `document.getElementById()`
- Reading and updating visible text using `innerText`
- Creating elements using `document.createElement()`
- Adding elements using `appendChild()`
- Removing elements using `remove()`
- Performing basic DOM CRUD operations

Allowed DOM Features

Use only the following DOM features in this assignment:

- `document.getElementById()`
- `element.innerText`
- `document.createElement()`
- `parent.appendChild()`
- `element.remove()`

Do not use:

- `querySelector()`
- `querySelectorAll()`
- `textContent`
- `innerHTML`

- `append()`
- `classList`
- Events or `addEventListener()`

Required Files

Create a folder named:

```
javascript-dom-assignment
```

Inside it, create:

```
index.html  
script.js
```

Connect `script.js` near the end of the `body` in `index.html`.

General Instructions

- Write all JavaScript inside `script.js`.
- Use meaningful variable names.
- Use `const` when the variable will not be reassigned.
- Run the page using Live Server.
- Check the browser page and console after every task.
- Do not use DOM methods that were not taught in class.
- Add comments before every task in `script.js`.

Task 1: Select and Read a Heading

Create a heading in HTML with the ID `mainTitle`.

Requirements:

- Select the heading using `document.getElementById()`.
- Print the selected element in the console.

- Print the heading text using `innerText`.

Task 2: Update Page Title

Create an `h1` element with the text:

Old JavaScript Title

Requirements:

- Give it the ID `pageTitle`.
- Select it using `getElementById()`.
- Update its text to:

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Task 3: Update Student Details

Create the following HTML elements:

- Student name
- Student course
- Student city

Give each element a unique ID.

Requirements:

- Select all three elements individually.
- Update their text using `innerText`.
- Use your own student details as the new values.

Task 4: Create and Add a Paragraph

Create an empty `div` with the ID `messageContainer`.

Requirements:

- Select the container.

- Create a new `p` element using `document.createElement()`.
- Set its text to:

```
DOM element created successfully
```

- Add the paragraph inside the container using `appendChild()`.

Task 5: Create a Student List Item

Create an empty `ul` with the ID `studentList`.

Requirements:

- Create one `li` element.
- Add a student name using `innerText`.
- Append the `li` inside the `ul`.

Expected page structure:

```
Student List
└─ One student name
```

Task 6: Add Multiple Students

Use the same `studentList` from Task 5.

Requirements:

- Create three separate `li` elements.
- Add a different student name to each element.
- Append all three elements to the list.
- Do not write the student names directly inside the HTML list.

Task 7: Create Course Cards

Create an empty `div` with the ID `courseContainer`.

Create three course sections using JavaScript.

Each course section must contain:

- One heading element for the course name
- One paragraph element for the trainer name

Suggested courses:

- JavaScript
- Python
- FastAPI

Requirements:

- Create all elements using `createElement()`.
- Set their content using `innerText`.
- Append the heading and paragraph to their course container.
- Append every course container to `courseContainer`.

Task 8: Update a Created Element

Create a paragraph using JavaScript with the initial text:

```
Course registration is closed
```

Append it to the page.

After appending it, update the same paragraph text to:

```
Course registration is open
```

Task 9: Remove an Existing HTML Element

Create this element directly in HTML:

```
<p id="oldMessage">This message should be removed</p>
```

Requirements:

- Select it using `getElementById()`.

- Remove it using `remove()`.
- Confirm that it is no longer visible on the page.

Task 10: Remove a Dynamically Created Element

Create and append a paragraph using JavaScript.

Use this text:

Temporary notification

After appending it:

- Print the element in the console.
- Remove the same element using `remove()`.
- Confirm that it no longer appears on the page.

Task 11: DOM CRUD Practice

Create an empty `div` with the ID `studentContainer`.

Perform all four CRUD operations:

Create

Create a paragraph containing a student name and append it to the container.

Read

Print the paragraph text using `innerText`.

Update

Change the student name to another name using `innerText`.

Delete

Remove the paragraph using `remove()`.

Task 12: Product List Challenge

Create an empty `ul` with the ID `productList`.

Using JavaScript, add these products:

- Laptop
- Mobile
- Keyboard
- Mouse

Requirements:

- Create one `li` element for each product.
- Use `innerText` for the product name.
- Use `appendChild()` to add every item.
- Update `Keyboard` to `Mechanical Keyboard`.
- Remove the `Mouse` item.

Final visible list:

```
Laptop
Mobile
Mechanical Keyboard
```

Main Challenge: Student Profile Builder

Create an empty `div` with the ID `profileContainer`.

Build a complete student profile using only JavaScript.

The profile must contain:

- Student name as an `h2`
- Course as a `p`
- Age as a `p`
- City as a `p`
- Status as a `p`

Initial values:

```
Name: Rahul  
Course: JavaScript  
Age: 21  
City: Hyderabad  
Status: Learning
```

After creating and appending the profile:

- Update the course to `Full Stack Development`.
- Update the status to `Course Completed`.
- Remove the age element.

Final visible profile:

```
Rahul  
Full Stack Development  
Hyderabad  
Course Completed
```

Bonus Challenge: Classroom Notice Board

Create an empty `div` with the ID `noticeBoard`.

Using JavaScript:

- Create a heading named `Classroom Notices`.
- Add four notice paragraphs.
- Update the second notice.
- Remove the fourth notice.
- Append all remaining elements to the notice board.

Do not write notice content directly in the HTML file.